MNXB11 Project:
Lund vs Umeå Climate Comparison

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1 Introduction

The aim of this project is to compare the climate of Lund and Umeå using historical meteorological data from the Swedish Meteorological and Hydrological Institute (SMHI). The data are processed using Bash scripts and C++, while ROOT is used for statistical analysis and visualisation.

The project investigates four aspects of the climate of the two cities.

The first analysis examines seasonal variation in mean temperature. It investigates how mean temperature varies throughout the year in Lund and Umeå and how their seasonal temperature patterns differ.

The second analysis investigates long-term changes in mean annual temperature. It examines how annual mean temperature has changed over time in the two cities and whether observable long-term trends can be identified.

The third analysis focuses on precipitation. It investigates how precipitation varies throughout the year and between the two cities, including which city receives more precipitation during different parts of the year.

The fourth analysis examines observed global solar irradiance. It investigates how incoming solar radiation varies throughout the year in Lund and Umeå and how the difference in latitude between the cities affects the observed irradiance.

Together, these analyses provide complementary perspectives on the climate of Lund and Umeå. The report describes the data processing and analysis methods used, presents the resulting figures, and discusses the findings in relation to the project's research questions.

1.1 Code collaboration and version control

The project uses Git and GitHub for collaborative development. We chose collaboration policy A, in which the release manager maintains the main upstream repository while the other team members develop changes in their own forks.

Each substantial contribution is developed on a separate branch and published to the developer's remote repository. The changes are then submitted to the upstream repository through a pull request. The release manager reviews the proposed changes and decides whether they should be merged into the main branch.

This approach separates development from integration and allows contributions to be reviewed before they become part of the main project. It also provides a traceable history of changes through Git commits and pull requests.

The project repository is available at:
`https://github.com/Manuel0516/MNXB11-Climate-Comparison`

1.2 Implementation and execution

The project was implemented using Bash, C++, and ROOT. Bash scripts were used to prepare and organize the input data and coordinate the execution of the analysis programs. C++ was used to process the meteorological measurements and perform the calculations, while ROOT was used to generate the plots presented in this report.

The project was run using an Apptainer container provided for the MNXB11 course on the COSMOS cluster.

A SLURM batch job was also configured to run the project on COSMOS. The batch script specified the required resources and launched the common execution script, allowing the analyses to run through the cluster's job scheduler.

2 Variation in Daily Mean Temperature Analysis

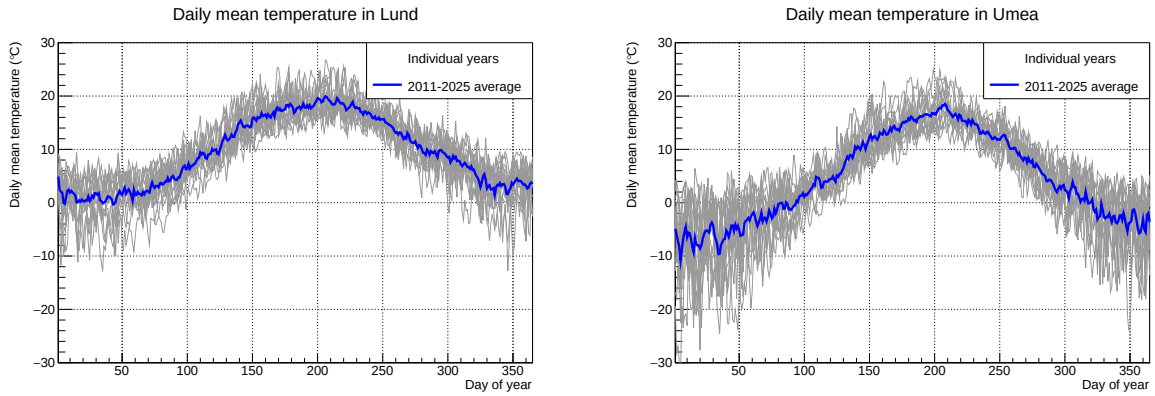
2.1 Method

This analysis compares daily mean temperatures in Lund and Umeå over the period 2011–2025. Daily mean temperatures were calculated from the available observations. A daily mean was included only when at least two numerical observations were available for that day, avoiding single-measurement daily means. Both G (good-quality) and Y (estimated or interpolated) observations were included.

To compare the seasonal patterns across years, the data were represented using a common 365-day calendar. February 29 was excluded from leap years, and the day indices after February were adjusted accordingly. This ensures that the same seasonal day is compared across all years.

For each city, the daily temperature curves for individual years are plotted in grey, while the mean seasonal temperature across the available years is shown in blue. A third figure shows the mean seasonal temperature in Lund and Umeå together to allow an easier comparison.

2.2 Results



(a) Lund (b) Umeå

Figure 1: Daily mean temperatures for individual years from 2011 to 2025 (grey) and the mean seasonal temperature (blue) in (a) Lund and (b) Umeå.

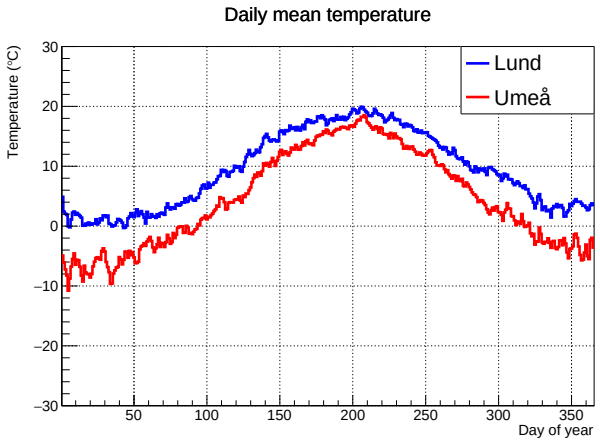


Figure 2: Comparison of mean daily temperature in Lund and Umeå over the period 2011–2025.

2.3 Discussion

Figure 1 shows the daily mean temperature for each year from 2011 to 2025, together with the mean seasonal temperature for each city. Both cities exhibit a clear seasonal cycle, with temperatures increasing from winter towards summer and decreasing again during autumn. The individual-year curves also reveal noticeable variability around the mean seasonal profile. This variability is particularly apparent during winter, when temperatures fluctuate considerably between years.

Figure 2 directly compares the mean seasonal temperatures in Lund and Umeå. Lund is generally warmer throughout the year, and the difference is particularly pronounced during winter, when Umeå has substantially lower mean temperatures. During summer, the two curves approach each other, although Lund remains generally warmer. Both cities reach their highest mean temperatures around the middle of the year, followed by a decrease towards winter.

Overall, the results reveal a clear seasonal temperature cycle in both locations, with warmer conditions in Lund and a larger contrast between the cities during the colder part of the year.

3 Long-term Changes in Mean Annual Temperature Analysis

3.1 Method

The original temperature dataset was cleaned using a bash script, removing metadata and data points that were deemed inaccurate with possible extrapolation errors. This derived dataset was utilised by together with the previous analysis. Mean annual temperature per year as well as plotting histograms and fitting functions were done in C++ and ROOT. To parse the CSV into C++, a common parser function was used throughout the project. Calculations, plotting and fitting were done specific to each analysis, hence individually.

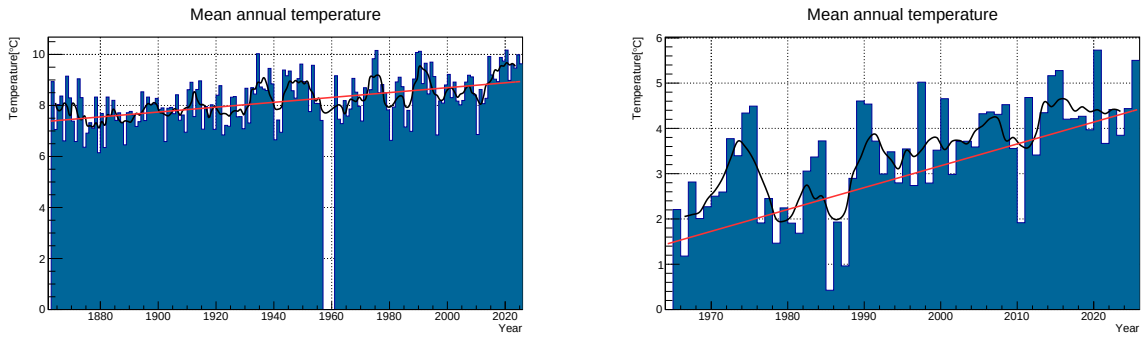
The Lund dataset was much larger than the Umeå dataset which began consistent data collection in 1965, however was missing data from years 1958 to 1961 as well as logging an incomplete year for 1957. The year was therefore removed due to systematic error stemming from the inclusion of an incomplete year. By the same argument, data for the current year, 2026 was also omitted as the entire year has not elapsed.

Despite both datasets consisting of a similar number of valid measurements across each day, the Umeå dataset included estimated interpolations for times between valid measurements whereas Lund only consisted of only valid measurements. Since we are dealing with the mean temperature across years, all interpolated data points were kept in the dataset as due to the law of large numbers, as our number of observations increase, our sample mean will better reflect the true mean of the data. Furthermore, any interpolated points which are possibly incorrect would have minor impact on results as they would be averaged out by the numerous other data points within the year. The mean annual temperature T_y was calculated by,

$$T_y = \frac{1}{N(y)} \sum_{i=0}^{N(y)} t_i,$$

where y represents the year, $N(y)$, the number of observations in year y and the ith temperature observation of the year represented by t_i. Lastly, after plotting the histogram of mean annual temperatures, both a moving average function and a first order polynomial were fitted to visualise long-term trends.

3.2 Results



(a) Lund (b) Umeå

Figure 3: Mean annual temperature for Lund(1863-2025) and Umeå(1965-2025)

Station	p_0	Error	p_1	Error
Lund	-10.4001	± 9.26532	0.00954816	± 0.00477083
Umeå	-93.2776	± 25.488	0.0482257	± 0.0128032
Lund (from 1965)	-26.347	± 42.9346	0.0175226	± 0.0215216

Table 1: Parameters for first order polynomial fit, $y = p_0 + p_1 \cdot x$

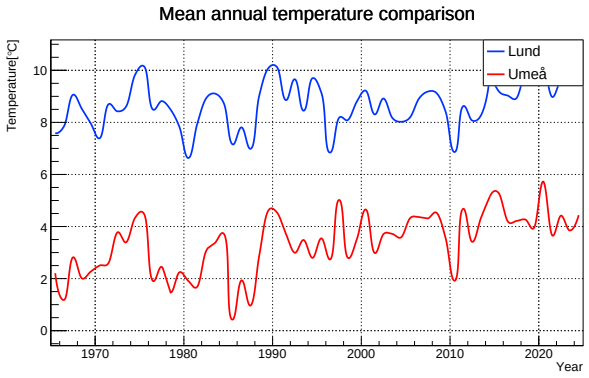


Figure 4: Comparison of mean annual temperature in Lund vs Umeå over the period 1965-2025.

Figure 3 visualizes the mean annual temperature trend for individual stations across their data collection lifetime. As mentioned in the method, we can see the cavity in Lund's dataset on the interval 1957 to 1961. From the figure we can see that in comparison to Lund, Umeå possesses a higher average temperature due to its position further north which can be seen more clearly in figure 4. Both Lund and Umeå shows a consistent increase in mean annual temperatures as time progresses. We can see from table 1, the p_1 gradient values for Lund and Umeå are positive despite taking into account the uncertainties.

Comparing the trend lines for Lund pre- and post-1965 from table 1, we see that the gradient is increasing with time, indicating to us that the rate of mean annual temperature increase is also increasing. Secondly, comparing gradients between Lund and Umeå post-1965, we can see that Umeå has a greater rate of change relative to Lund, which could be explained by polar amplification and the ice-albedo effect.

Similarly, looking at the error values for the same p_1, we can see that compared to the longer Lund trend line, there is a much higher uncertainty. This could be explained statistically due to a lesser number of observations in a shorter timeline, but it could also be an indication of more frequent and/or more extreme/volatile weather conditions which could represent the effects of global warming for example. However, this data alone is insufficient to viably draw a conclusion such as this.

3.3 Discussion

As stated earlier, all interpolated and estimated points were kept in the dataset and used as data which was supported by the law of large numbers and the fact that it mitigated data points that would be detrimental if the dataset were smaller. However on the flip side, accepting these less valid points also lead to some consequences. The loss of variance is both a strength as mentioned previously, but can also act negatively.

In cases where this data would be used to infer conclusions on global warming or extreme weather, the addition of interpolated points i.e the smoothing of the curves, would result in these extreme values being undervalued or seen as less extreme than in reality. However, when simply using the data to draw a conclusion on long term temperature trends, this smoothing or loss of variance should not affect our conclusion as real, valid data points were taken at consistent intervals throughout each day which would make any large incorrect interpolations unlikely.

4 Precipitation Comparison Analysis

4.1 Method

The seasonal daily mean precipitation in both Lund and Umeå from SMHI stations 53430 and 140490 respectively, was analysed[1, 2]. Data stretching from 1863 to 2026 was used for Lund, and 1956 to 2026 for Umeå. It was first preprocessed using bash to only include the relevant information, i.e. the date, the precipitation and the quality of the observation. A function, **extract_precipitation_data()**, was then implemented to extract the data into essentially a 2D list that could be handled by C++. More precisely, a class **PrecipitationMeasurement** was defined to represent each measurement, and a vector of those was used to represent the data. Another function, **calculate_day_stats()**, was then implemented to use this extracted data and perform statistical analysis on it such that each day of the year between January 1st and December 31st (1-365), excluding February 29th, would get a corresponding mean and standard deviation. The mean, μ, for each day was calculated by simply summing up the precipitation amounts for the day across the years and dividing it by the number of years. The standard deviation used was the population standard deviation

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}} \tag{1}$$

where x_i was each precipitation value and N the total population of values. The obtained stats were plotted using the TH1D class in ROOT which allows for the creation of histograms. Specifically, a TH1D object was created on the heap, and a for loop was implemented to set the contents and errors of the histogram bins - this content being the stats obtained from **calculate_day_stats()**.

4.2 Results

The results obtained are shown in Figure 5, using both confirmed good quality data ("G") and unconfirmed data ("Y"). Excluding the unconfirmed data gives the same qualitative result.

4.3 Discussion

Looking at the results it is clear that there is a pattern showing more rain in the second half of the year than in the first part of the year for both cities. The error bars show a similar pattern, indicating that we get the most activity around late summer and decreasing slowly until the winter. Comparing both cities though, it appears Umeå has a larger variance in the autumn, suggesting more variance in how much it rains.

One clear issue with these results is that the error is very significant compared to the mean values, giving a pretty high uncertainty. The reason for this is likely due to the fact that it doesn't rain on a lot of days. It would be interesting to compare the precipitation levels on only the days that it does rain, although that would be a different analysis. One could also look at the monthly or weekly patterns to get a better idea of the seasonal variation. Lastly, one could also refine the method of calculating the errors, like using the sample standard deviation instead, although that specifically wouldn't make a significant difference.

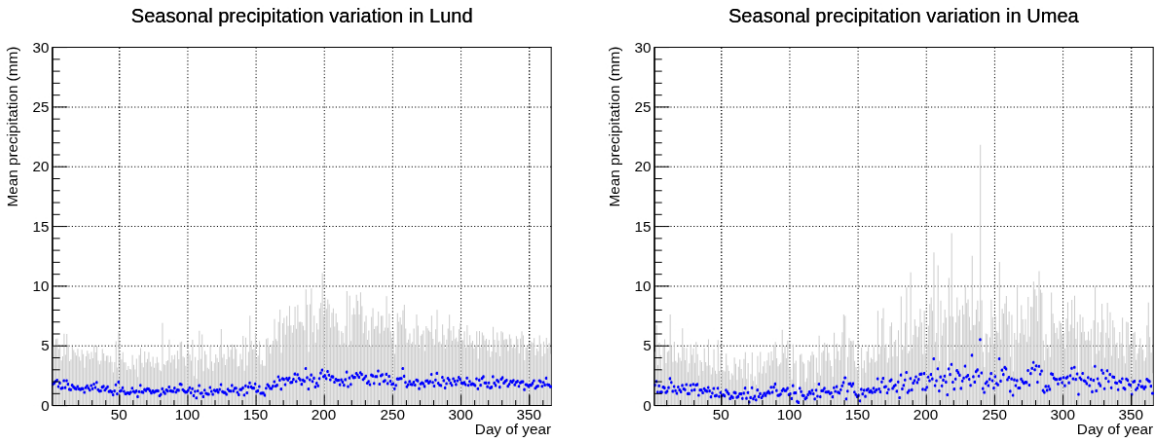


Figure 5: Seasonal precipitation variation for both Lund and Umeå with daily means and standard deviation as error.

5 Observed Solar Irradiance Analysis

5.1 Methods

Hourly solar irradiance data were obtained from the Swedish Meteorological and Hydrological Institute (SMHI), using the Lund Sol (station 53445) and Umeå Sol (station 140615) measurement stations [3, 4].

The dataset consisted of hourly solar irradiance measurements from SMHI for Lund and Umeå, including quality flags indicating valid measurements ("G") or interpolated/estimated values ("Y"). A Bash script was used to remove unnecessary metadata, retaining only the date, time, irradiance, and quality columns. Data quality analysis and calculations were performed in C++.

For Lund, only 304 out of 381294 observations were flagged as estimated, representing approximately 0.08% of the dataset. This fraction was considered too small to significantly affect the overall conclusions. Furthermore, most estimated values occurred on separate days, typically affecting only a single hour, further limiting their influence on daily averages. The only notable exception occurred in 2024, with two days containing more than five estimated hourly measurements. Nevertheless, these values were retained, as their expected influence on the final results was negligible. Similarly, Umeå contained 144 estimated observations out of 381294 (0.04%), with no days exceeding five estimated measurements. No observations were removed or modified based on their quality flags.

To ensure consistent calendar-day comparisons across years, February 29 was excluded, and subsequent day indices in leap years were shifted by one. Given the limited contribution of a single winter day to annual irradiation, this simplification was unlikely to affect the overall conclusions significantly.

Hourly mean irradiance values ($H_{y,d,h}$), expressed in $\mathrm{W/m^2}$, were added over each day to obtain daily irradiation. Here, y, d, and h denote the year, day of the year, and hour, respectively. The resulting daily irradiation is expressed in $\mathrm{Wh/m^2}$:

$$D_{y,d} = \sum_{h=0}^{23} H_{y,d,h} \Delta t, \qquad \Delta t = 1\,\mathrm{h} \tag{2}$$

Daily irradiation was then averaged across corresponding calendar days to characterize the seasonal distribution while reducing annual fluctuations:

$$\overline{D}_d = \frac{1}{N_d} \sum_{y \in Y_d} D_{y,d} \tag{3}$$

where Y_d denotes the set of years containing observations for day d, and N_d is the corresponding number of years.

The annual mean irradiance was calculated by dividing the total annual irradiation by the number of hours in the year:

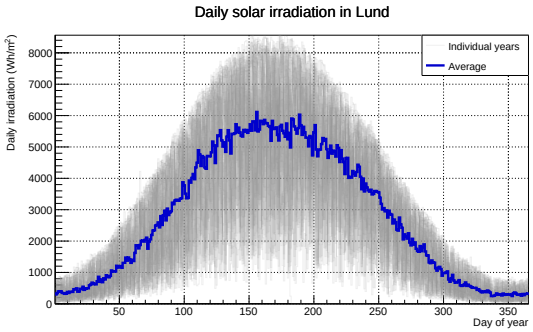
$$\overline{I}_y = \frac{1}{T_y} \sum_{d=1}^{365} D_{y,d} \tag{4}$$

where $T_y = 8760\,\mathrm{h}$ for complete years, keeping the exclusion of February 29. The resulting annual mean irradiance is expressed in $\mathrm{W/m^2}$.

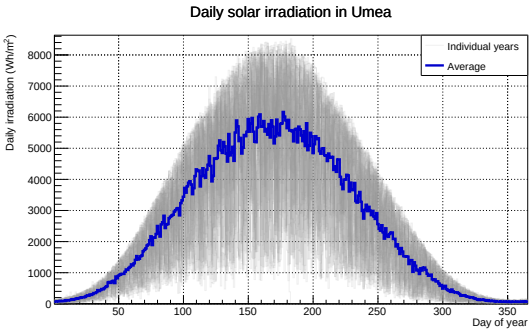
The final year (2026) was excluded from annual mean calculations, as measurements were only available through the sunnier part of the year. Including this incomplete year would introduce a systematic upward bias, potentially leading to misleading conclusions regarding annual variability and long-term trends.

Finally, a first-order polynomial was fitted to the annual mean irradiance values to estimate the long-term temporal trend.

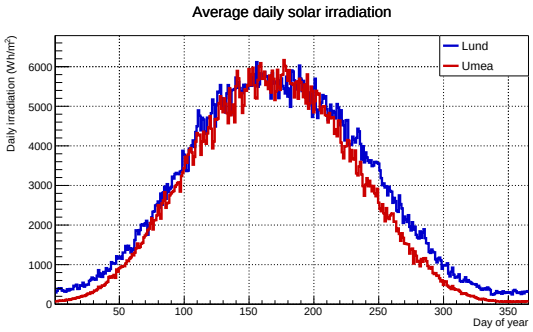
5.2 Results



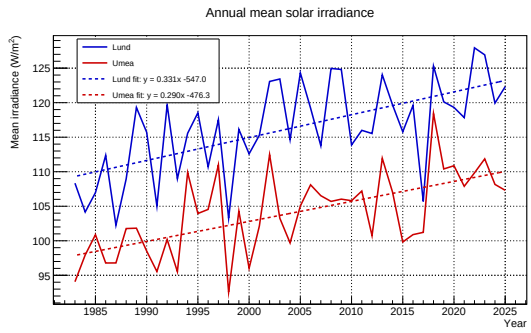
(a) Daily solar irradiation in Lund.



(b) Daily solar irradiation in Umeå.



(c) Comparison of average daily irradiation.



(d) Annual mean irradiance and linear fits.

Figure 6: Solar irradiance analysis for Lund and Umeå. (a–b) Daily irradiation for individual years and the average seasonal profile. (c) Comparison of average daily irradiation. (d) Annual mean irradiance with linear regression fits.

5.3 Discussion

Figures 6a and 6b show the season effect on solar irradiation, with maxima irradiation during summer and minima during winter. Umeå exhibits a more pronounced seasonal contrast due to its higher latitude, resulting in lower irradiance during winter days. During the summer, the different latitud angle gets compensated by longer summer daylight hours which makes the average irradiance between Lund and Umea very similar during summer month as can be observed in 6c. The individual yearly measurements show a lot variability due to shadows and clouds, which is reduced when averaging over the full period. Lund receives more irradiation throughout the winter, although summer maxima are comparable.

Figure 6d shows an increasing trend in annual mean irradiance for both cities, with linear regression slopes of approximately $0.33\,\mathrm{W\,m^{-2}\,yr^{-1}}$ for Lund and $0.29\,\mathrm{W\,m^{-2}\,yr^{-1}}$ for Umeå. Despite these positive trends, significant year-to-year fluctuations remain, likely associated with variations in cloud cover. A particularly large increase is observed in 2018, consistent with SMHI reports of an exceptionally sunny spring and summer that year [5]. In the final year (2025), Lund shows a modest increase, while Umeå experiences a slight decrease.

6 General Discussion and Conclusions

The comparison showed clear seasonal patterns in temperature, precipitation, and solar irradiance for both Lund and Umeå.

Lund generally experiences higher temperatures throughout the year, with the largest differences occurring during winter. Similarly, Umeå receives considerably less solar irradiation during winter, while longer daylight hours compensate for its lower solar elevation during summer, resulting in comparable irradiation between the cities.

Precipitation tends to be higher during the second half of the year in both cities. However, the large standard deviations indicate substantial variability, making precise comparisons between individual days difficult.

The annual temperature and solar irradiance analyses both show positive long-term fitted trends. Nevertheless, considerable year-to-year variability remains, and the different observation periods and uncertainties must be considered when comparing the cities.

Overall, the results illustrate how geographical location influences seasonal climate patterns, while also demonstrating the importance of long-term observations and statistical averaging when analysing large quantities of data.

7 Use of Generative Artificial Intelligence

Generative artificial intelligence tools, including ChatGPT, were used as supplementary support during the development of this project. These tools were used to assist with writing, reviewing and understanding parts of the C++ implementation, as well as troubleshooting errors encountered during compilation and execution.

References

[1] SMHI. Daily precipitation observations – lund. `https://www.smhi.se/data/hitta-data-for-en-plats/ladda-ner-vaderobservationer/precipitation24HourSum/53430`. Accessed: 2026-10-31.

[2] SMHI. Daily precipitation observations – umeå. `https://www.smhi.se/data/hitta-data-for-en-plats/ladda-ner-vaderobservationer/precipitation24HourSum/140490`. Accessed: 2026-10-31.

[3] SMHI. Solar irradiance observations – lund sol. https://www.smhi.se/data/hitta-data-for-en-plats/ladda-ner-vaderobservationer/irradiance/53445. Accessed: 2026-10-31.

[4] SMHI. Solar irradiance observations – umeå sol. https://www.smhi.se/data/hitta-data-for-en-plats/ladda-ner-vaderobservationer/irradiance/140615. Accessed: 2026-10-31.

[5] Swedish Meteorological and Hydrological Institute. Solstrålning i sverige sedan 1983. `https://www.smhi.se/kunskapsbanken/meteorologi/stralning/solstralning-i-sverige-sedan-1983`. Accessed: 2026-10-28.